

PDMP

Rethinking Balanced Multimodal Learning via Performance-Dominant Modality Prioritization

Research Paper Summary with Key Figures and Tables

Paper Focus	Balanced / Imbalanced Multimodal Learning
Main Idea	Prioritize the modality with stronger unimodal performance instead of forcing equal balance.
Core Method	Performance-Dominant Modality Prioritization (PDMP)
Relevance to Project	Useful for audio + facial-expression prototypes where one modality may be more informative or reliable.

1. Paper Overview

This paper studies the under-optimization problem in multimodal learning. In many multimodal systems, the final model does not fully exploit all available modalities and may even perform worse than a strong unimodal model. The key argument of the paper is that forcing all modalities to learn equally is not always optimal. Instead, the modality that performs best when trained alone should be identified and prioritized during multimodal training.

Main Thesis

Balanced learning is not always the best strategy. In some cases, an intentionally guided imbalance led by the performance-dominant modality can achieve better multimodal performance.

2. Problem Statement

Previous multimodal imbalance methods usually assume that weak modalities should be strengthened and dominant modalities should be suppressed. However, this assumption can be misleading because some modalities naturally contain more useful information for a specific task. For example, in one dataset the visual modality may be more informative than audio, while in another dataset audio may be more reliable. Therefore, forcing audio and visual features to contribute equally may reduce the model performance.

Traditional View	PDMP View
Imbalance is always harmful.	A guided imbalance can be useful when led by the most informative modality.
Weak modalities should be emphasized.	The performance-dominant modality should receive priority.
The goal is balanced optimization.	The goal is better performance and stronger information use.

Figure 1. Balanced vs. Imbalanced Learning

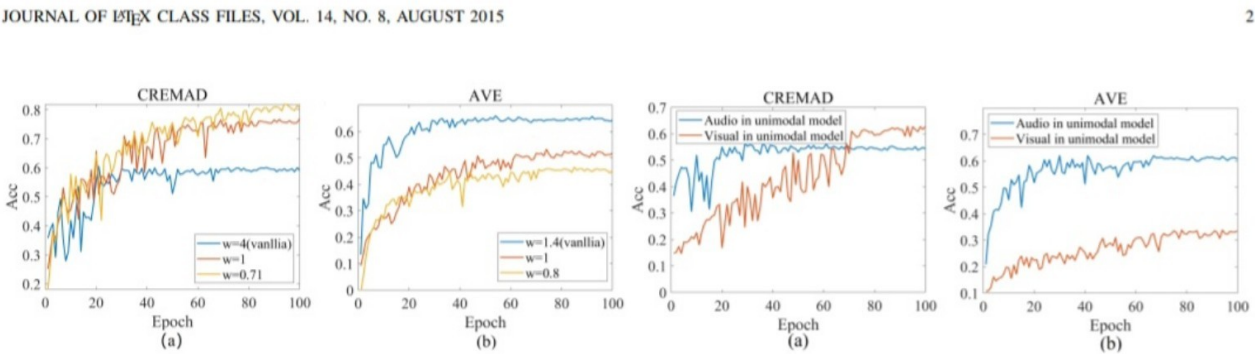


Figure 1: Performance curves on CREMA-D and AVE. The results show that equal balance is not always optimal; the best performance can occur under an imbalanced setting.

The figure supports the core claim of the paper. On CREMA-D, the best performance is not obtained at the balanced point. On AVE, forcing balance may reduce performance. This motivates the need to identify which modality should guide optimization.

3. Proposed Method: PDMP

The proposed method is called Performance-Dominant Modality Prioritization (PDMP). It is a plug-and-play strategy that can be applied to different multimodal architectures and fusion methods.

3.1 Modality Analysis

PDMP first trains a unimodal model for each input modality. Then, it ranks the modalities according to their individual unimodal performance. The modality with the best unimodal performance is defined as the performance-dominant modality.

- Audio-only model is trained and evaluated.
- Visual-only model is trained and evaluated.
- The modality with the higher unimodal score is selected as the performance-dominant modality.

3.2 Gradient Modulation

After identifying the performance-dominant modality, PDMP uses asymmetric gradient modulation. The gradient of the performance-dominant modality is increased so that it can guide the multimodal optimization process.

- The performance-dominant modality receives stronger optimization priority.
- The remaining modality is adjusted according to whether it is also optimization-dominant.
- The goal is not equal learning, but performance-oriented modality prioritization.

Figure 2. PDMP Architecture and Gradient Modulation

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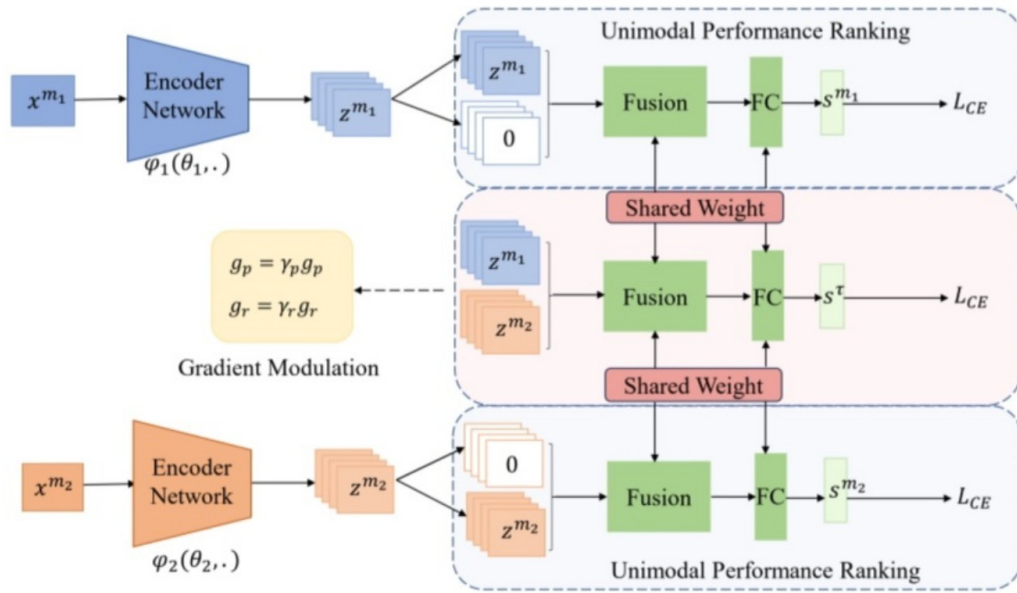


Figure 2: PDMP identifies the performance-dominant modality using unimodal ranking and then modulates gradients to make this modality guide the multimodal training process.

4. Modality Ranking and Training Procedure

The paper emphasizes that the modality that dominates inside a multimodal model is not necessarily the modality with the best standalone performance. Therefore, unimodal analysis is necessary before applying gradient modulation.

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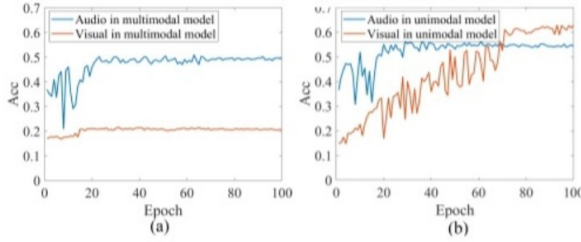


Fig. 3. The visualization on the CREMA-D datasets. (a) presents the performance of the audio and visual branches in the multimodal model. (b) presents the performance of the audio and visual modality in the unimodal model.

unimodal model for each modality and rank their performance. According to the definition, the modality with better performance at the end of training is the performance-dominant modality. Besides, according to the memorization effect [41]–[43], that PDMP tend to first memorize simple samples

Algorithm 1 Multimodal learning with PDMP strategy

Require: Training dataset D , iteration number T , hyperparameter γ_p .

Modality Analysis:

Train the unimodal model for each input modality within T iterations;

Determine the performance-dominant modality and the remaining modality via unimodal performance ranking;

Determine optimization-dominant modality via modality branch performance ranking in the multimodal;

Determine the parameter γ_r via Eq.12;

Gradient Modulation:

for $t = 0, \dots, T - 1$ **do**

 Feed-forward the batched data to the model.

 Calculate backward gradient.

 Modify the gradient of each modality via Eq.10 and

11;

 Update model parameters via the modified gradient.

Figure 3 and Algorithm 1: The figure compares unimodal and multimodal modality behavior, while the algorithm summarizes the PDMP training procedure.

Algorithm 1 can be summarized in two stages: first, analyze modality performance using unimodal models; second, perform multimodal training with gradient modulation that prioritizes the performance-dominant modality.

5. Experimental Results

5.1 Comparison with Existing Modulation Strategies

TABLE I
COMPARISON WITH EXISTING MODULATION STRATEGIES ON CREMA-D, KINETICS-SOUNDS, AND AVE DATASETS. THE PROPOSED PDMP ACHIEVES THE BEST PERFORMANCE. BOLD AND UNDERLINE MEAN THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY.

Dataset	CREMA-D		KS		AVE	
	Acc	macro F1	Acc	macro F1	Acc	macro F1
Audio-only	57.27	57.89	48.67	48.89	62.16	58.54
Visual-only	62.17	62.78	52.36	52.67	31.40	29.87
Concatenation	58.83	59.43	64.97	65.21	66.15	62.46
Grad-Blending	68.81	69.34	67.31	67.68	67.40	63.87
OGM-GE	64.34	64.93	66.35	66.76	65.62	62.97
AGM	67.21	68.04	65.61	65.99	64.50	61.49
PMR	65.12	65.91	65.01	65.13	63.62	60.36
MMPareto	70.19	70.82	69.13	69.05	68.22	64.54
MLA	73.21	73.77	<u>69.62</u>	<u>69.98</u>	<u>70.92</u>	<u>67.23</u>
D&R	<u>73.52</u>	<u>73.96</u>	69.10	69.36	69.62	64.93
PDMP	80.21	80.34	72.18	72.67	71.28	67.49

Table I: PDMP achieves the best performance on CREMA-D, Kinetics-Sounds (KS), and AVE datasets compared with existing modulation and balancing methods.

Table I demonstrates that PDMP outperforms strong baselines such as Grad-Blending, OGM-GE, AGM, PMR, MMPareto, MLA, and D&R. This validates the claim that performance-dominant prioritization can improve multimodal learning more effectively than simple balancing.

5.2 Generalization to Three Modalities and Large Datasets

TABLE II
(LEFT): COMPARISON WITH IMBALANCED MULTIMODAL LEARNING METHODS ON CEFA DATASET WITH THREE MODALITIES. (RIGHT):
COMPARISON WITH IMBALANCED MULTIMODAL LEARNING METHODS ON THE UCF101 DATASET WITH 101 CATEGORIES. BOLD AND UNDERLINE
MEAN THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY.

Dataset	CEFA		UCF101	
Methods	Acc	macro F1	Acc	macro F1
Concatenation	64.91	65.26	80.41	79.40
Grad-Blending	65.68	66.11	81.73	80.84
OGM-GE	66.33	66.76	81.15	80.36
AGM	67.65	67.98	81.55	80.36
PMR	68.38	68.89	81.36	80.37
MMPareto	71.18	71.61	81.98	80.64
MLA	71.78	72.02	82.01	<u>81.22</u>
D&R	<u>72.66</u>	<u>72.97</u>	<u>82.11</u>	80.87
PDMP	74.45	74.78	83.65	82.08

Table II: PDMP also performs best on CEFA with three modalities and UCF101 with 101 action categories.

The CEFA experiment is important because it includes three modalities: RGB, Depth, and IR. The results show that PDMP is not limited to two-modality scenarios. The UCF101 results further show that PDMP remains effective on a larger action recognition dataset.

5.3 Compatibility with Different Fusion Methods

TABLE III
PERFORMANCE ON CREMA-D, AVE AND KS DATASETS WITH VARIOUS
FUSION METHODS. COMBINED WITH PDMP, CONVENTIONAL FUSION
METHODS CONSISTENTLY GAIN CONSIDERABLE IMPROVEMENT. †
INDICATES PDMP IS APPLIED.

Dataset	CREMA-D	AVE	KS
Method	Acc	Acc	Acc
Audio-only	57.27	62.16	48.67
Visual-only	62.17	31.40	52.36
Concatenation	58.83	66.15	64.97
Summation	62.24	67.25	64.53
Film	56.21	59.18	57.35
Gated	57.72	65.21	63.28
Concatenation†	80.21	71.28	72.18
Summation†	82.54	71.65	73.17
Film†	70.21	64.19	67.89
Gated†	80.18	71.15	73.96

Table III: Applying PDMP improves several conventional fusion methods, including Concatenation, Summation, FiLM, and Gated fusion.

These results indicate that PDMP is flexible. It is not tied to a specific fusion method and can be applied as a general optimization strategy for multimodal learning.

5.4 Optimization Dependency Analysis

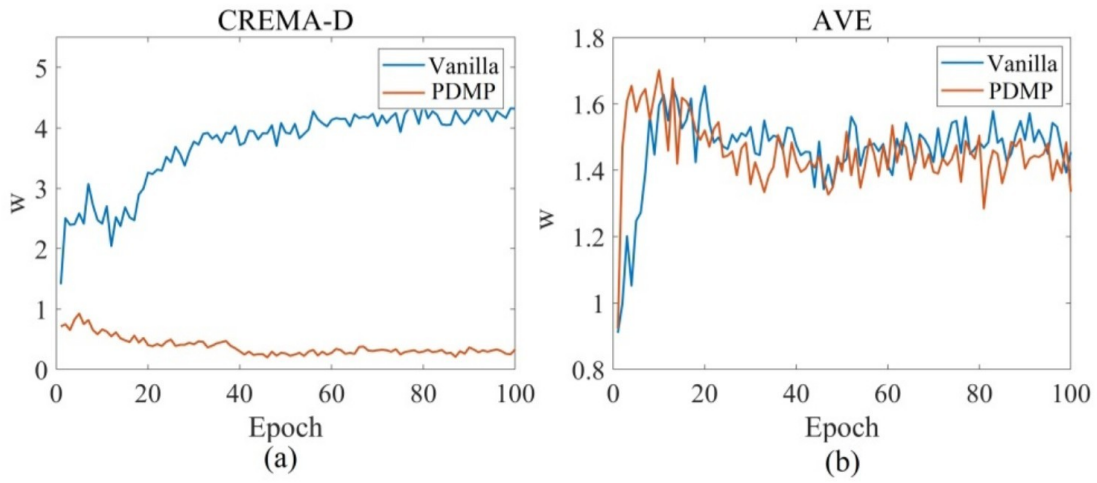


Figure 4: PDMP changes the optimization dependency coefficient w so that the performance-dominant modality can guide optimization more appropriately.

This figure shows how PDMP changes the model dependency during training. On CREMA-D, PDMP reduces excessive dependency on the wrong modality. On AVE, it preserves the dominance of the modality that is already performance-dominant.

6. Relevance to Our Multimodal CIR Project

This paper is highly relevant to our project because our prototype will combine audio and facial-expression features. In real-world conditions, audio and visual signals are not always equally reliable.

- Audio may be noisy, silent, or distorted.
- Facial expressions may be unclear due to blur, lighting, occlusion, or pose variation.
- A fixed equal-weight fusion strategy may fail when one modality is more informative than the other.

PDMP suggests that a practical multimodal prototype should first estimate which modality is more informative for the task and then prioritize it during training or decision-making.

7. Research Gap

Research Gap

Although PDMP improves multimodal learning by prioritizing the performance-dominant modality, it mainly focuses on training-stage optimization. In real-world audio-visual emotion recognition systems, modality reliability may change dynamically due to noisy audio, blurred faces, missing frames, or changing lighting conditions. Therefore, applying performance-dominant prioritization in a dynamic real-world prototype remains an important practical gap.

8. Key Takeaways

- Balanced learning is not always optimal in multimodal learning.
- The strongest unimodal modality may need to guide multimodal optimization.
- PDMP identifies the performance-dominant modality through unimodal performance ranking.
- PDMP uses asymmetric gradient modulation to prioritize that modality.
- The method achieves strong results across multiple datasets, fusion methods, and architectures.
- For our prototype, the idea can help decide whether audio or facial expressions should be trusted more under different conditions.

9. One-Sentence Summary

Final Summary

PDMP argues that multimodal learning should not always force equal balance between modalities; instead, it should identify the modality with stronger unimodal performance and prioritize it during training through asymmetric gradient modulation.